CSE-1223
Discrete Mathematics
Segment 7: Graph

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Graph

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Graph coloring

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Graph

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1

Graph terminology

Graph terminology

What is a **graph**? [Aut'17]

A *graph* $G = (V, E)$ consists of V, a nonempty set of *vertices* (or *nodes*) and E, a set of *edges*. Each edge has either one or two vertices associated with it, called its *endpoints*. An edge is said to *connect* its endpoints.

Graph terminology

Write the definition of **digraph**.

A *directed graph* (or *digraph*) (V, E) consists of a nonempty set of vertices V and a set of *directed edges* (or *arcs*) E. Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to *start* at u and *end* at v.

Graph terminology

What is the **neighborhood** of a vertex?

The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called the *neighborhood* of v. If A is a subset of V, we denote by $N(A)$ the set of all vertices in G that are adjacent to at least one vertex in A. So, $N(A) = \bigcup_{v \in A} N(v)$.

Example: What are the neighborhoods of the vertices in the graph G?

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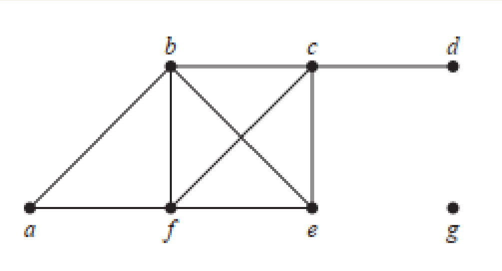

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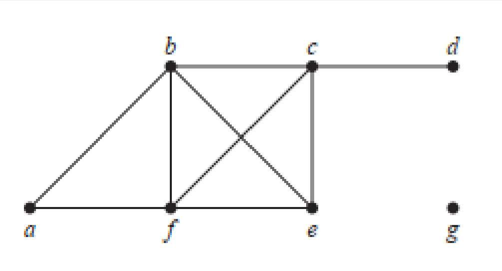

The neighborhoods of these vertices are
$N(a) = \{b, f\}$, $N(b) = \{a, c, e, f\}$, $N(c) = \{b, d, e, f\}$,
$N(d) = \{c\}$, $N(e) = \{b, c, f\}$, $N(f) = \{a, b, c, e\}$,
and $N(g) = \emptyset$.

Graph terminology

What is the **degree** of a vertex?

The *degree of a vertex in an undirected graph* is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

Example: What are the degrees of the vertices in the graph G?

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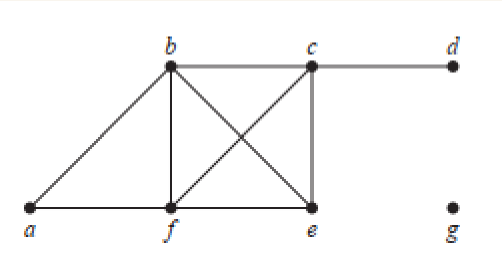

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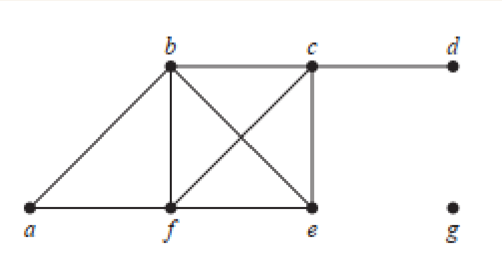


In G,
deg(a) = 2, deg(b) = deg(c) = deg(f) = 4,
deg(d) = 1, deg(e) = 3, and
deg(g) = 0.

Graph terminology

State the **Handshaking Theorem**? [**Spr'18**]

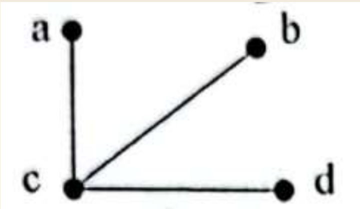
THE HANDSHAKING THEOREM Let $G = (V, E)$ be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} \deg(v).$$

(Note that this applies even if multiple edges and loops are present.)

Graph terminology

Example: Verify the Handshaking theorem for the following figure. [**Spr'18**]



Solution:

deg(a) = deg(b) = deg(d) = 1,

deg(c) = 3

Now, 2m = 2*3 = 6

deg(a) + deg(b) + deg(d) + deg(c) = 1 +1 +1 +3 = 6

$$2m = \sum_{v \in V} \deg(v).$$

Graph terminology

Example: How many edges are there in a graph with 10 vertices each of degree six?

Solution:

Because the sum of the degrees of the vertices is $6 \cdot 10 = 60$, it follows that

$2m = 60$

where m is the number of edges.

Therefore, $m = 30$.

$$2m = \sum_{v \in V} \deg(v).$$

Graph terminology

Example: Proof that An undirected graph has an even number of vertices of odd degree. [**Aut'17**]

Proof: Let V_1 be the vertices of even degree and V_2 be the vertices of odd degree in an undirected graph $G = (V, E)$ with m edges. Then

even → $$2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

must be even since $\deg(v)$ is even for each $v \in V_1$

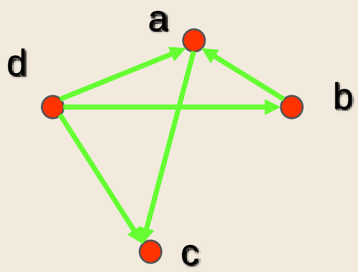
This sum must be even because $2m$ is even and the sum of the degrees of the vertices of even degrees is also even. Because this is the sum of the degrees of all vertices of odd degree in the graph, there must be an even number of such vertices.

Graph terminology

What are the **initial vertex** and **terminal vertex** of a directed edges?

Ans:

When (u, v) is an edge of the graph G with directed edges, u is said to be *adjacent to* v and v is said to be *adjacent from* u. The vertex u is called the *initial vertex* of (u, v), and v is called the *terminal* or *end vertex* of (u, v). The initial vertex and terminal vertex of a loop are the same.



Initial Vertex	Terminal Vertices
a	c
b	a
c	
d	a, b, c

Graph terminology

What are the **in-degree** and **out-degree** of a vertex?

Ans:

In a graph with directed edges the *in-degree of a vertex v*, denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex. The *out-degree of v*, denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex. (Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)

Example: Find the in-degree and out-degree of each vertex in the graph G?

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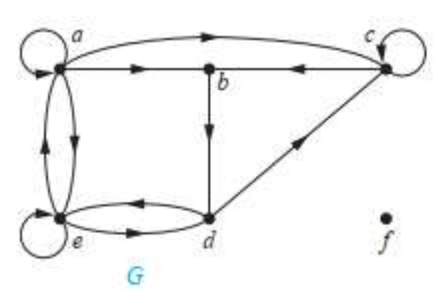

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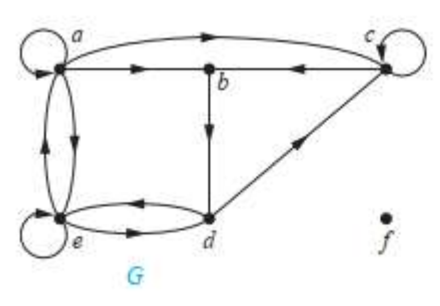
Example: Find the in-degree and out-degree of each vertex in the graph G?

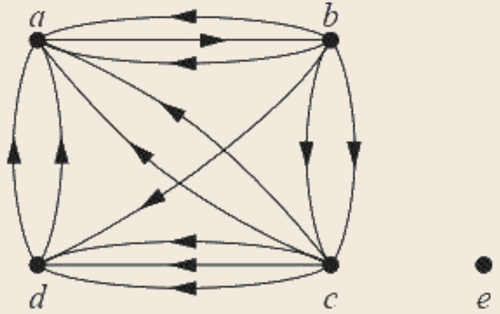


The in-degrees in G are
deg−(a) = 2, deg−(b) = 2, deg−(c) = 3, deg−(d) = 2,
deg−(e) = 3, and deg−(f) = 0.
The out-degrees are deg+(a) = 4, deg+(b) = 1, deg+(c) = 2, deg+(d) = 2, deg+(e) = 3, and deg+(f) = 0

Graph terminology

Example: Determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the given directed multigraph. [**Aut'16**]

Solution:



This directed multigraph has 5 vertices and 13 edges. The in-degree of vertex a is deg- (a) = 6 since there are 6 edges with a as their terminal vertex; its out-degree is deg+(a) = 1.

Similarly we have deg-(b) = 1, deg+(b) = 5, deg-(c) = 2, deg+(c) = 5, deg-(d) = 4, deg+(d) = 2, deg-(e) = 0, and deg+(e) = 0 (vertex e is isolated). As a check we see that the sum of the in-degrees and the sum of the out-degrees are both equal to the number of edges (13).

Graph terminology

Prove that the sum of the in-degrees and the sum of the out-degrees of a directed graph are the same. [**Aut’16**]

Ans:

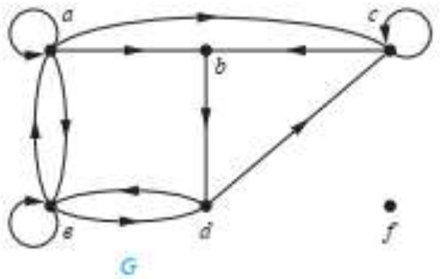


FIGURE 2 **The Directed Graph *G*.**

Solution: The in-degrees in G are $\deg^-(a) = 2$, $\deg^-(b) = 2$, $\deg^-(c) = 3$, $\deg^-(d) = 2$, $\deg^-(e) = 3$, and $\deg^-(f) = 0$. The out-degrees are $\deg^+(a) = 4$, $\deg^+(b) = 1$, $\deg^+(c) = 2$, $\deg^+(d) = 2$, $\deg^+(e) = 3$, and $\deg^+(f) = 0$.

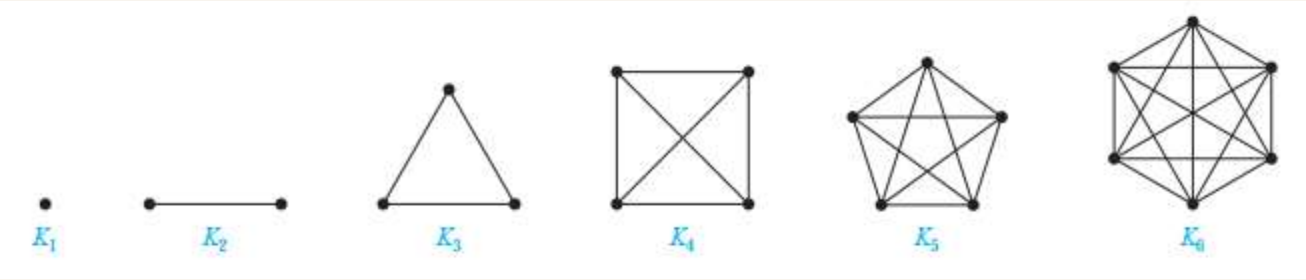
Because each edge has an initial vertex and a terminal vertex, the sum of the in-degrees and the sum of the out-degrees of all vertices in a graph with directed edges are the same. Both of these sums are the number of edges in the graph. This result is stated as Theorem 3.

Graph terminology

Define **Complete graph**. [**Spr'18,23,'24**]

Ans:

Complete Graphs A **complete graph on n vertices**, denoted by K_n, is a simple graph that contains exactly one edge between each pair of distinct vertices. The graphs K_n, for $n = 1, 2, 3, 4, 5, 6$, are displayed in Figure 3. A simple graph for which there is at least one pair of distinct vertex not connected by an edge is called **noncomplete**.



For k_3 degree sequence (2,2,2)

[**Spr'23**] [**Spr'24**]

Graph terminology

Define **Cycles**.

Ans:

Cycles A **cycle** C_n, $n \geq 3$, consists of n vertices $v_1, v_2, \ldots, v_n$ and edges $\{v_1, v_2\}$, $\{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$. The cycles C_3, C_4, C_5, and C_6 are displayed in Figure 4.

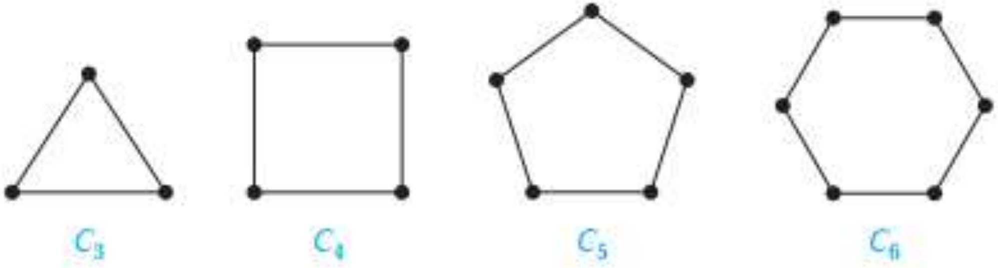


FIGURE 4 **The Cycles C_3, C_4, C_5, and C_6.**

Graph terminology

Define **Wheels**.

Ans:

Wheels We obtain a **wheel** W_n when we add an additional vertex to a cycle C_n, for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n, by new edges. The wheels W_3, W_4, W_5, and W_6 are displayed in Figure 5.

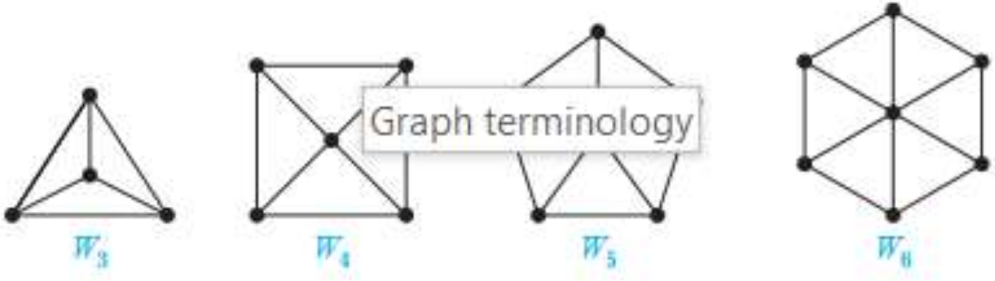


FIGURE 5 The Wheels W_3, W_4, W_5, and W_6.

Graph terminology

Define the **Bipartite graph**. [**Spr'18,23, Aut'17**]

Ans:

A simple graph G is called *bipartite* if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a *bipartition* of the vertex set V of G.

Graph terminology

Define **Complete Bipartite Graph**. [**Spr'18,'24**]

Ans:

Complete Bipartite Graphs **A complete bipartite graph $K_{m,n}$** is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset. The complete bipartite graphs $K_{2,3}$, $K_{3,3}$, $K_{3,5}$, and $K_{2,6}$ are displayed in Figure 9.

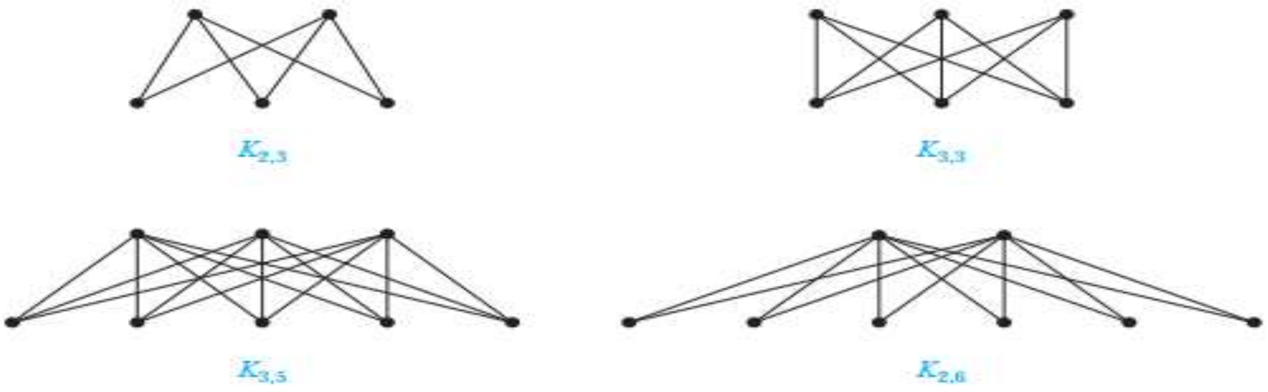


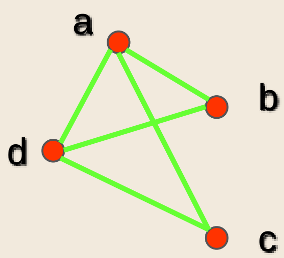
FIGURE 9 **Some Complete Bipartite Graphs.** [**Spr'24**]

2

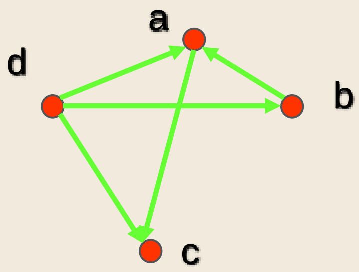
Representation of graph

Representing Graphs

Adjacency lists specify the vertices that are adjacent to each vertex of the graph.



Vertex	Adjacent Vertices
a	b, c, d
b	a, d
c	a, d
d	a, b, c



Initial Vertex	Terminal Vertices
a	c
b	a
c	
d	a, b, c

Representing Graphs

Definition: Let $G = (V, E)$ be a simple graph with $|V| = n$. Suppose that the vertices of G are listed in arbitrary order as $v_1, v_2, \ldots, v_n$.

The **adjacency matrix** A (or A_G) of G, with respect to this listing of the vertices, is the $n \times n$ zero-one matrix with 1 as its (i, j) entry when v_i and v_j are adjacent, and 0 otherwise.

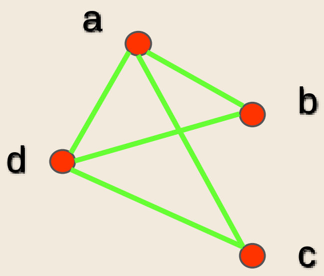
In other words, for an adjacency matrix $A = [a_{ij}]$,

$a_{ij} = 1$ if $\{v_i\ v_j\}$ is an edge of G,

$a_{ij} = 0$ otherwise.

Representing Graphs

Example: What is the adjacency matrix AG for the following graph G based on the order of vertices a, b, c, d ?

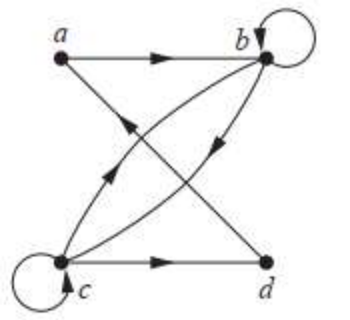


Solution:

$$A_G = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Representing Graphs

Example: What is the adjacency matrix AG for the following directed graph G based on the order of vertices a, b, c, d ?



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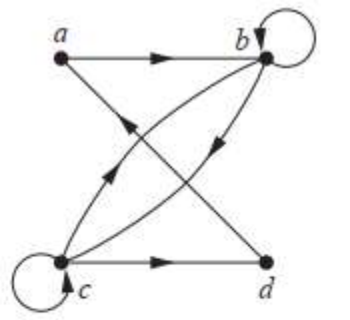

Solution:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

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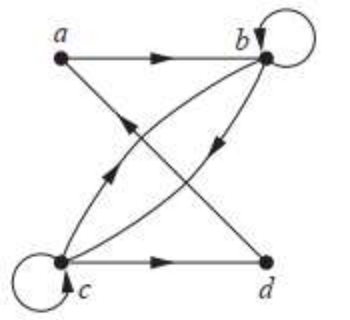
Example: What is the adjacency matrix AG for the following directed graph G based on the order of vertices a, b, c, d ?



Solution:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Exercise: Draw the directed graph by the following adjacency matrix. [**Spr'18**]

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

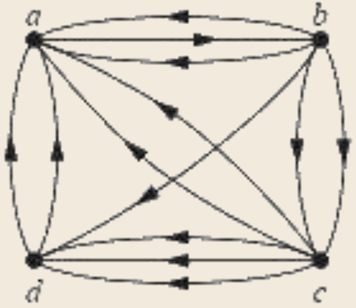
Representing Graphs

EXAMPLE 5 Use an adjacency matrix to represent the pseudograph shown in Figure 5.

Solution: The adjacency matrix using the ordering of vertices a, b, c, d is

$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}.$$

Exercise: Find the adjacency matrix of the following directed multigraph. [**Spr’17**]



Representing Graphs

Definition: Let $G = (V, E)$ be an undirected graph with $|V| = n$. Suppose that the vertices and edges of G are listed in arbitrary order as $v_1, v_2, \ldots, v_n$ and $e_1, e_2, \ldots, e_m$, respectively.

The **incidence matrix** of G with respect to this listing of the vertices and edges is the $n \times m$ zero-one matrix with 1 as its (i, j) entry when edge e_j is incident with v_i, and 0 otherwise.

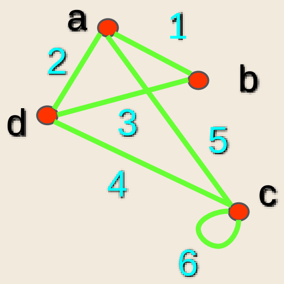
In other words, for an incidence matrix $M = [m_{ij}]$,

$m_{ij} = 1$ if edge e_j is incident with v_i

$m_{ij} = 0$ otherwise.

Representing Graphs

Example: What is the incidence matrix M for the following graph G based on the order of vertices a, b, c, d and edges 1, 2, 3, 4, 5, 6?

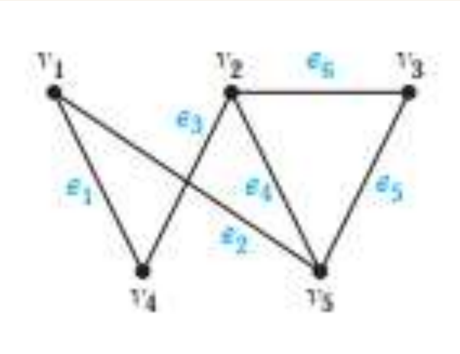


Solution:

$$M = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Representing Graphs

Example: Represent the graph shown in Figure with an incidence matrix.



Solution:

$$
\begin{array}{c}
\\
v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5
\end{array}
\begin{array}{c}
\begin{array}{cccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{array} \\
\begin{bmatrix}
1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 1 \\
1 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 1 & 0
\end{bmatrix}
\end{array}.
$$

3

Graph Isomorphism

Isomorphism of Graphs

What is **isomorphism?** [**Aut'17,Sp'23**]

Definition: The simple graphs $G_1 = (V_1, E_1)$ and G2 = (V_2, E_2) are **isomorphic** if there is a bijection (an one-to-one and onto function) f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if f(a) and f(b) are adjacent in G_2, for all a and b in V_1.

Such a function f is called an **isomorphism**.

In other words, G_1 and G_2 are isomorphic if their vertices can be ordered in such a way that the adjacency matrices M_{G_1} and M_{G_2} are identical.

Isomorphism of Graphs

From a visual standpoint, G1 and G2 are isomorphic if they can be arranged in such a way that their **displays are identical** (of course without changing adjacency).

Unfortunately, for two simple graphs, each with n vertices, there are **n! possible isomorphisms** that we have to check in order to show that these graphs are isomorphic.

However, showing that two graphs are **not** isomorphic can be easy.

Isomorphism of Graphs

For this purpose we can check **invariants**, that is, properties that two isomorphic simple graphs must both have.

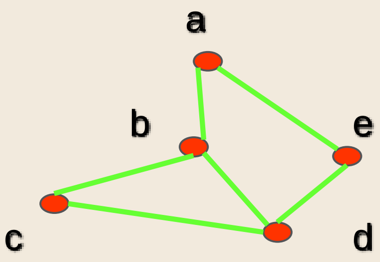
For example, they must have

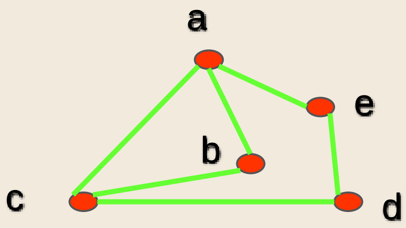
- the same number of **vertices**,
- the same number of **edges**, and
- the same **degrees** of vertices.

Note that two graphs that **differ** in any of these invariants are not isomorphic, but two graphs that **match** in all of them are not necessarily isomorphic.

Isomorphism of Graphs

Example: Are the following two graphs isomorphic?





Solution: Yes, they are isomorphic, because they can be arranged to look identical. You can see this if in the right graph you move vertex b to the left of the edge {a, c}. Then the isomorphism f from the left to the right graph is: f(a) = e, f(b) = a, f(c) = b, f(d) = c, f(e) = d.

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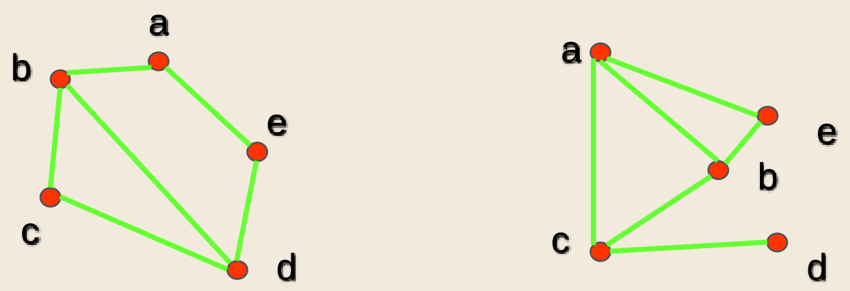
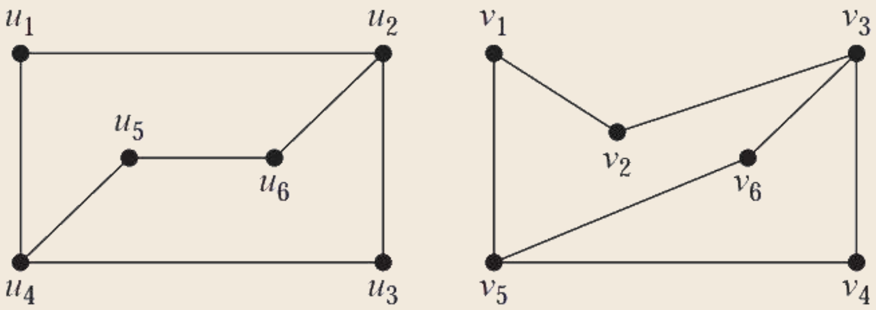


Solution: **No**, they are not isomorphic, because they differ in the degrees of their vertices. Vertex d in right graph is of degree one, but there is no such vertex in the left graph.

Isomorphism of Graphs

Example: Are the following two graphs isomorphic?



Solution: Both G and H have six vertices and seven edges. Both have four vertices of degree two and two vertices of degree three. It is also easy to see that the subgraphs of G and H consisting of all vertices of degree two and the edges connecting them are isomorphic (as the reader should verify). Because G and H agree with respect to these invariants, it is reasonable to try to find an isomorphism f .

4

Graph connectivity

Graph connectivity

Definition: A **path** of length n from u to v, where n is a positive integer, in an **undirected graph** is a sequence of edges $e_1, e_2, \ldots, e_n$ of the graph such that $e_1 = \{x_0, x_1\}, e_2 = \{x_1, x_2\}, \ldots, e_n = \{x_{n-1}, x_n\}$, where $x_0 = u$ and $x_n = v$.

When the graph is simple, we denote this path by its **vertex sequence** $x_0, x_1, \ldots, x_n$, since it uniquely determines the path.

The path is a **circuit** if it begins and ends at the same vertex, that is, if u = v.

The path or circuit is said to **pass through** or traverse x1, x2, ..., xn-1.

A path or circuit is **simple** if it does not contain the same edge more than once.

Graph connectivity

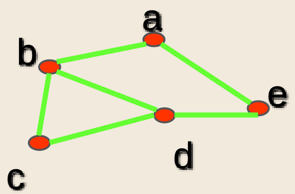
Definition: An undirected graph is called **connected** if there is a path between every pair of distinct vertices in the graph.

For example, any two computers in a network can communicate if and only if the graph of this network is connected.

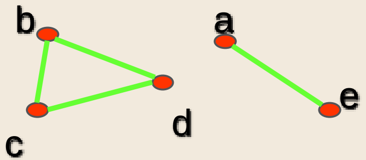
Note: A graph consisting of only one vertex is always connected, because it does not contain any pair of distinct vertices.

Graph connectivity

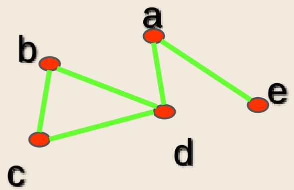
Example: Are the following graphs connected?



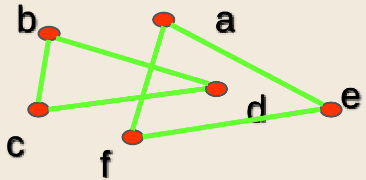
Yes



No



Yes



No

Graph connectivity

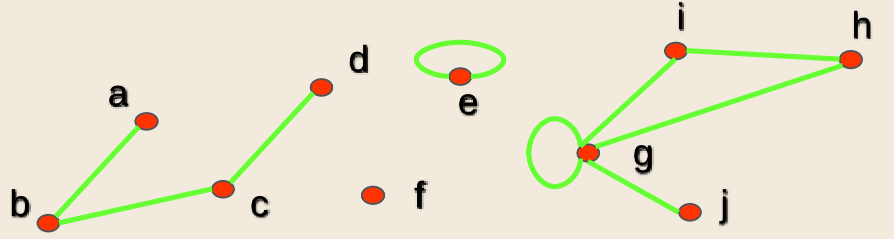
Definition: A graph that is not connected is the union of two or more connected subgraphs, each pair of which has no vertex in common. These disjoint connected subgraphs are called the **connected components** of the graph.

Definition: A **connected component** of a graph G is a maximal connected subgraph of G.

E.g., if vertex v in G belongs to a connected component, then all other vertices in G that is connected to v must also belong to that component.

Graph connectivity

Example: What are the connected components in the following graph?



Solution: The connected components are the graphs with vertices {a, b, c, d}, {e}, {f}, {i, g, h, j}.

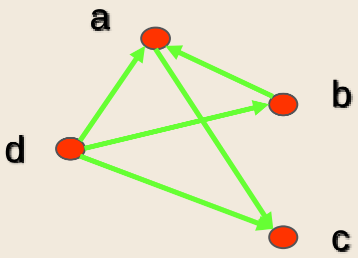
Graph connectivity

Definition: An directed graph is **strongly connected** if there is a path from a to b and from b to a whenever a and b are vertices in the graph.

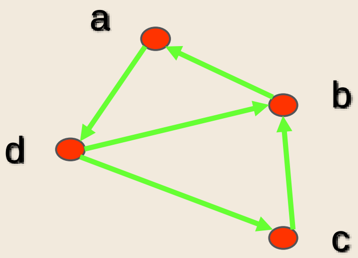
Definition: An directed graph is **weakly connected** if there is a path between any two vertices in the underlying undirected graph.

Graph connectivity

Example: Are the following directed graphs strongly or weakly connected?



Weakly connected, because, for example, there is no path from b to d.



Strongly connected, because there are paths between all possible pairs of vertices.

5

Planner Graphs

Planner Graphs

A graph is called **planar** if it can be drawn in the plane without any edges crossing (where a crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint). Such a drawing is called a **planar representation** of the graph

Planner Graphs

EXAMPLE 1 Is K_4 (shown in Figure 2 with two edges crossing) planar?

Solution: K_4 is planar because it can be drawn without crossings, as shown in Figure 3. ◀

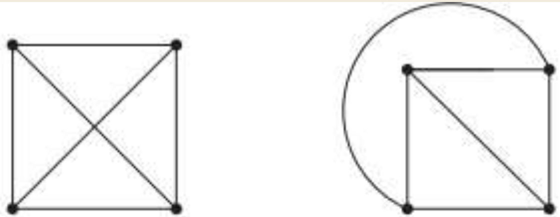


FIGURE 2 The Graph K_4.

FIGURE 3 K_4 Drawn with No Crossings.

Planner Graphs

EXAMPLE 2 Is Q_3, shown in Figure 4, planar?

Solution: Q_3 is planar, because it can be drawn without any edges crossing, as shown in Figure 5. ◀

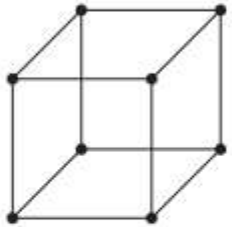


FIGURE 4 The Graph Q_3.

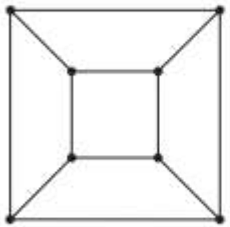
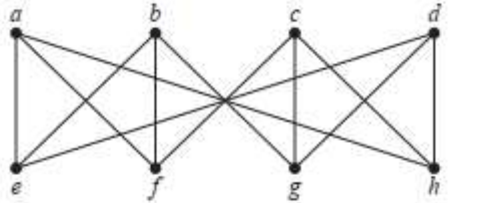


FIGURE 5 A Planar Representation of Q_3.

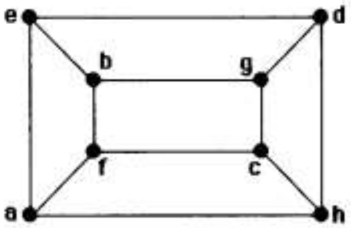
Planner Graphs

Use Kuratowski's theorem to determine whether the given graph is planar.



Solution:

The instructions are really not fair. It is hopeless to try to use Kuratowski's theorem to prove that a graph *is* planar, since we would have to check hundreds of cases to argue that there is no subgraph homeomorphic to K_5 or $K_{3,3}$. Thus we will show that this graph is planar simply by giving a planar representation. Note that it is Q_3.



6

Graph coloring

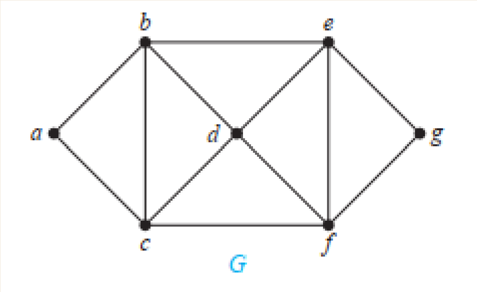
Graph coloring

A **coloring** of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

The **chromatic** number of a graph is the least number of colors needed for a coloring of this graph. The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter chi.)

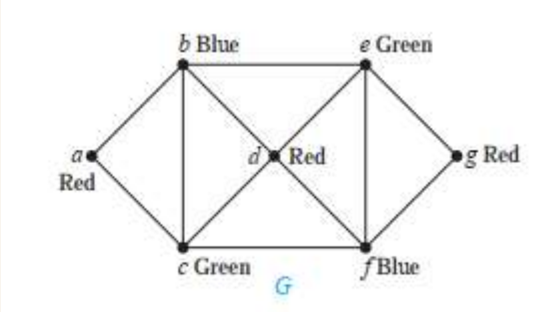
Graph coloring

THE FOUR COLOR THEOREM The chromatic number of a planar graph is no greater than four.



Example: What are the chromatic numbers of the graphs G.

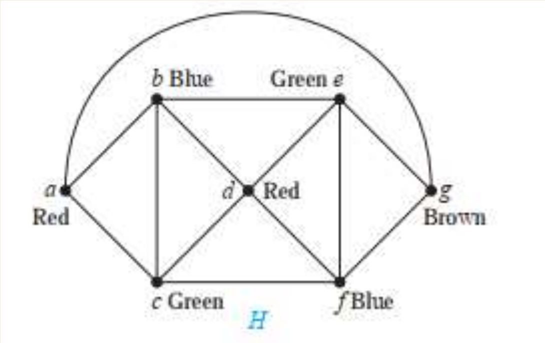
Solution: G using exactly three colors

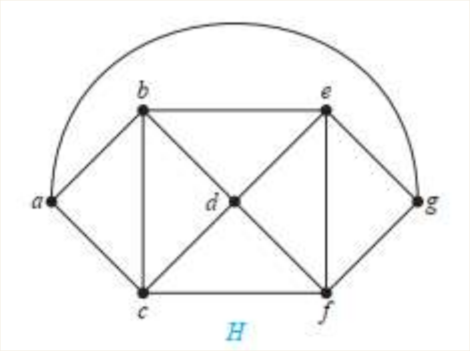


Graph coloring

Example: What are the chromatic numbers of the graphs H.

Solution: H has a chromatic number equal to 4.





Graph coloring

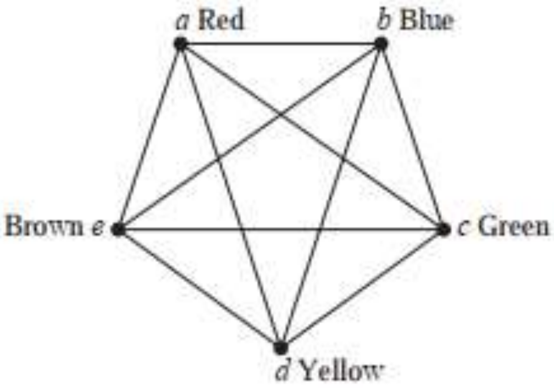


FIGURE 5 A Coloring of K_5.

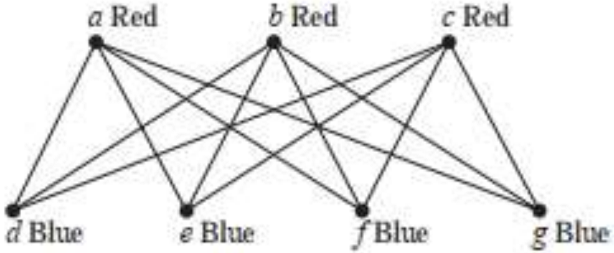


FIGURE 6 A Coloring of $K_{3,4}$.

Graph coloring

EXAMPLE 2 What is the chromatic number of K_n?

Solution: A coloring of K_n can be constructed using n colors by assigning a different color to each vertex. Is there a coloring using fewer colors? The answer is no. No two vertices can be assigned the same color, because every two vertices of this graph are adjacent. Hence, the chromatic number of K_n is n. That is, $\chi(K_n) = n$. (Recall that K_n is not planar when $n \geq 5$, so this result does not contradict the four color theorem.) A coloring of K_5 using five colors is shown in Figure 5. ◀

EXAMPLE 3 What is the chromatic number of the complete bipartite graph $K_{m,n}$, where m and n are positive integers?

Solution: The number of colors needed may seem to depend on m and n. However, as Theorem 4 in Section 10.2 tells us, only two colors are needed, because $K_{m,n}$ is a bipartite graph. Hence, $\chi(K_{m,n}) = 2$. This means that we can color the set of m vertices with one color and the set of n vertices with a second color. Because edges connect only a vertex from the set of m vertices and a vertex from the set of n vertices, no two adjacent vertices have the same color. A coloring of $K_{3,4}$ with two colors is displayed in Figure 6. ◀

Graph coloring

EXAMPLE 4 What is the chromatic number of the graph C_n, where $n \geq 3$? (Recall that C_n is the cycle with n vertices.)

Solution: We will first consider some individual cases. To begin, let $n = 6$. Pick a vertex and color it red. Proceed clockwise in the planar depiction of C_6 shown in Figure 7. It is necessary to assign a second color, say blue, to the next vertex reached. Continue in the clockwise direction; the third vertex can be colored red, the fourth vertex blue, and the fifth vertex red. Finally, the sixth vertex, which is adjacent to the first, can be colored blue. Hence, the chromatic number of C_6 is 2. Figure 7 displays the coloring constructed here.

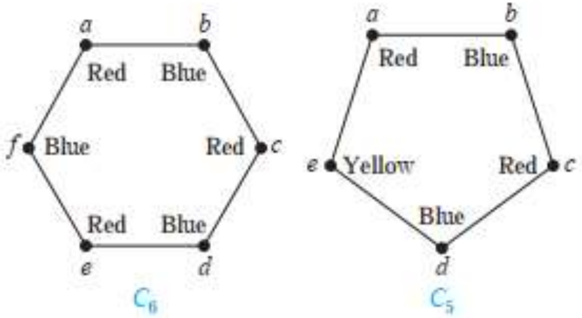


FIGURE 7 **Colorings of C_5 and C_6.**

7

Shortest Path Problems

Shortest Path Problems

- Shortest path problem is a problem of finding the shortest path(s) between vertices of a given graph.
- Shortest path between two vertices is a path that has the least cost as compared to all other existing paths.

Shortest Path Algorithms

- Shortest path algorithms are a family of algorithms used for solving the shortest path problem.

Applications:

Shortest path algorithms have a wide range of applications such as in-

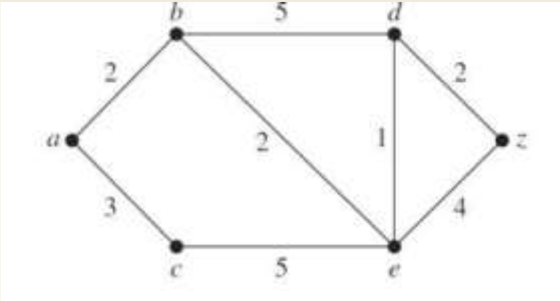
- Google Maps
- Road Networks
- Logistics Research

Dijkstra Algorithm

- Dijkstra Algorithm is a very famous greedy algorithm.
- It is used for solving the single source shortest path problem.
- It computes the shortest path from one particular source node to all other remaining nodes of the graph.

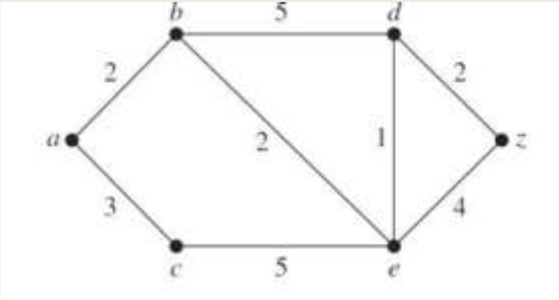
Dijkstra Algorithm

Find the shortest path from a to z of the following graph using Dijkstra's Algorithm.



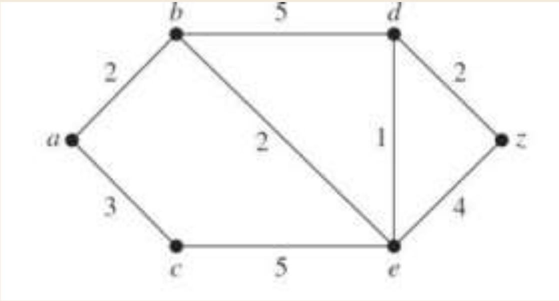
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)



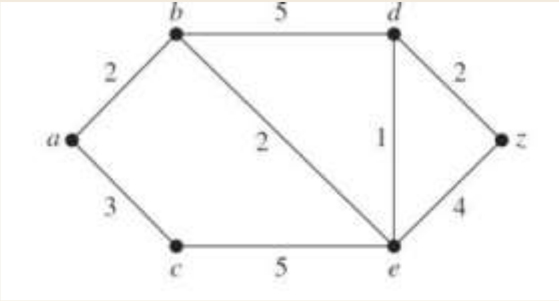
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞



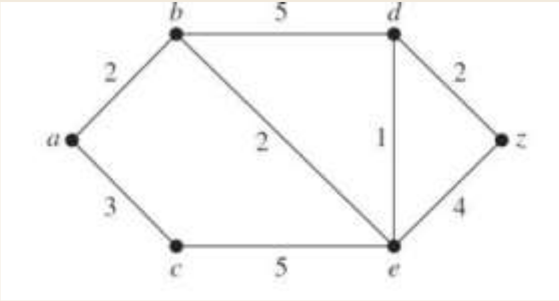
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞



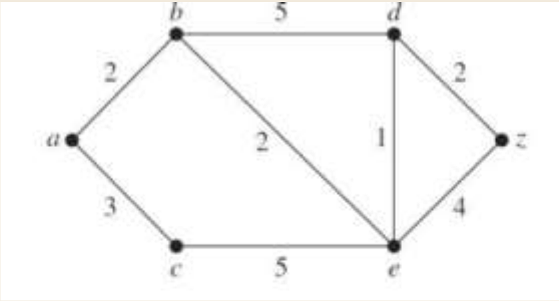
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab						



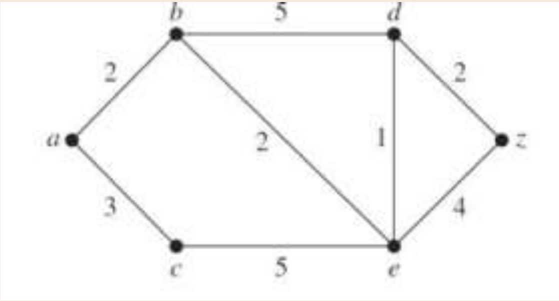
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞



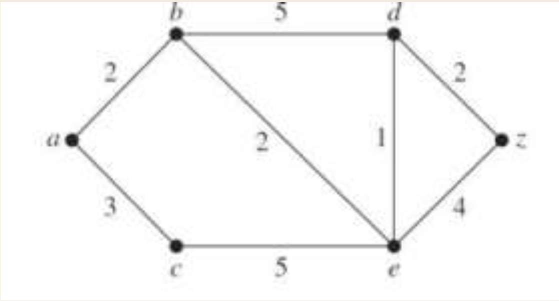
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞



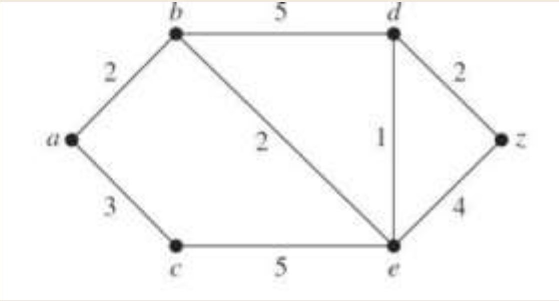
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc						



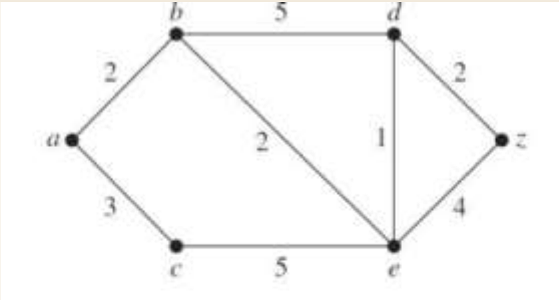
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞



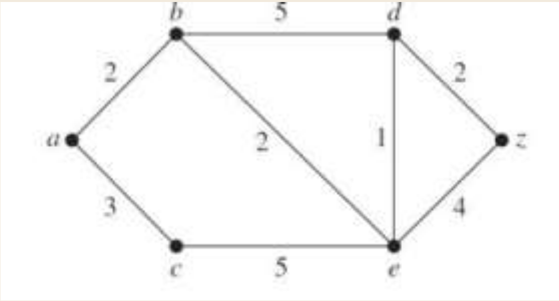
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞



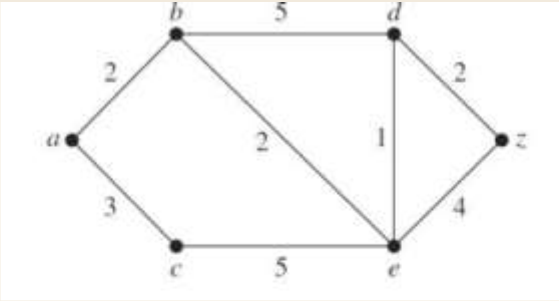
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞
4	abce						



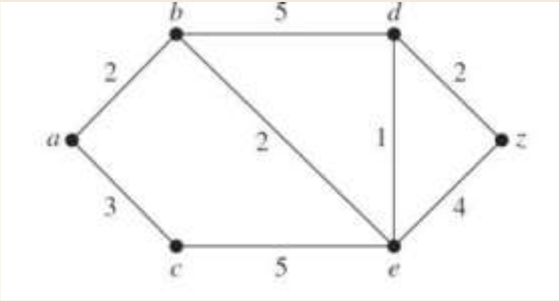
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞
4	abce				5	0	8



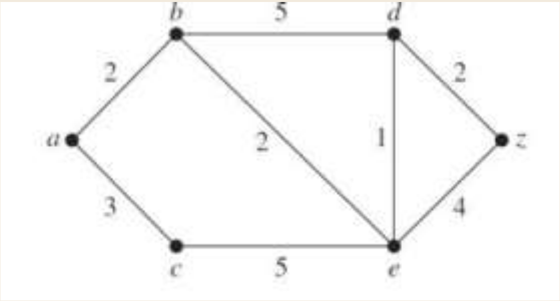
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞
4	abce				5	0	8



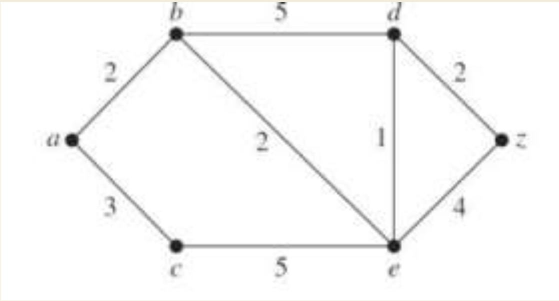
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞
4	abce				5	0	8
5	abced						



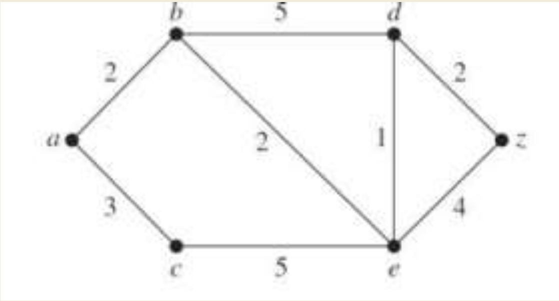
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞
4	abce				5	0	8
5	abced				0		7



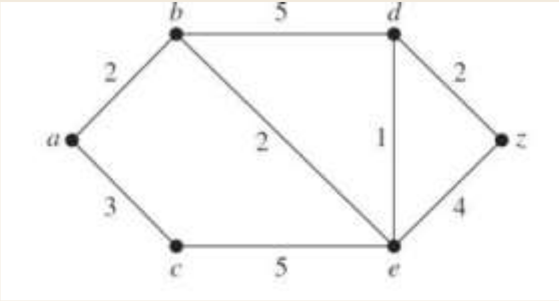
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞
4	abce				5	0	8
5	abced				0		7



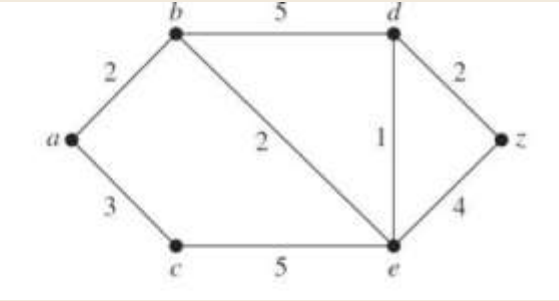
Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞
4	abce				5	0	8
5	abced				0		7
6	abcedz						0



Dijkstra Algorithm

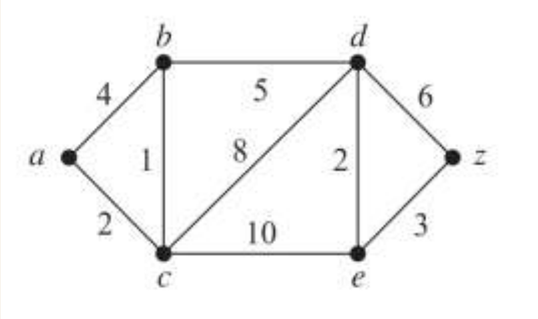
Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	2	3	∞	∞	∞
2	ab		0	3	7	4	∞
3	abc			0	7	4	∞
4	abce				5	0	8
5	abced				0		7
6	abcedz						0



Our shortest path is a=>b=>e=>d=>z with length 7.

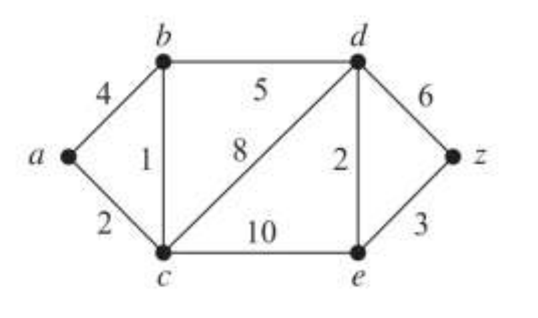
Dijkstra Algorithm

Find the shortest path from a to z of the following graph using Dijkstra's Algorithm.



Dijkstra Algorithm

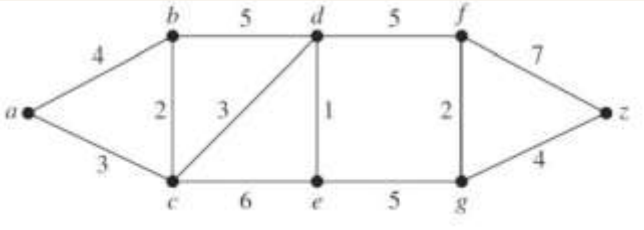
Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(z)
1	a	0	4	2	∞	∞	∞
2	ac		3	0	10	12	∞
3	acb		0		8	12	∞
4	acbd				0	10	14
5	acbde					0	13
6	acbdez						0



Our shortest path is a=>c=>b=>d=>e=>z with length 13

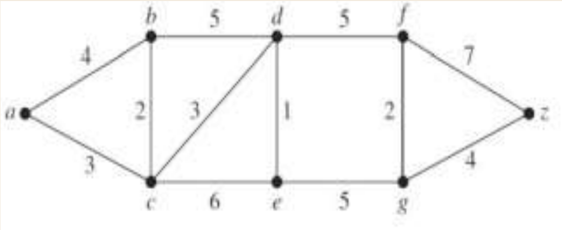
Dijkstra Algorithm

Find the shortest path from a to z of the following graph using Dijkstra's Algorithm.



Dijkstra Algorithm

Step	Node	D(a)	D(b)	D(c)	D(d)	D(e)	D(f)	D(g)	D(z)
1	a	0	4	3	∞	∞	∞	∞	∞
2	ac		4	0	6	9	∞	∞	∞
3	acb		0		6	9	∞	∞	∞
4	acbd				0	7	11	∞	∞
5	acbde					0	11	12	∞
6	acbdef						0	12	18
7	acbdefg							0	16
8	acbdefgz								



Our shortest path is a=>c=>d=>e=>g=>z with length 16

8

Euler and Hamilton paths

Euler paths & circuits

Definition: An **Euler circuit** in a graph G is a simple circuit containing every edge of G with no repeats. Being a circuit, it must start and end at the same vertex. An **Euler path** in G is a simple path containing every edge of G with no repeats. Being a path, it does not have to return to the starting vertex. [**Spr'16**]

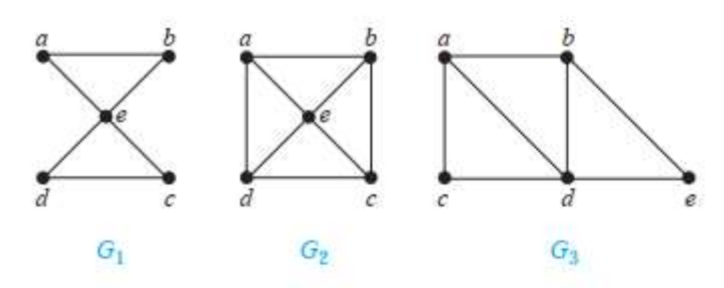
Euler paths & circuits

Theorem 1: A connected multigraph with at least two vertices has an Euler circuit if and only if each of its vertices has **even degree**.

Theorem 2: A connected multigraph has an Euler path but not an Euler circuit if and only if it has **exactly two** vertices of **odd degree**.

Euler paths & circuits

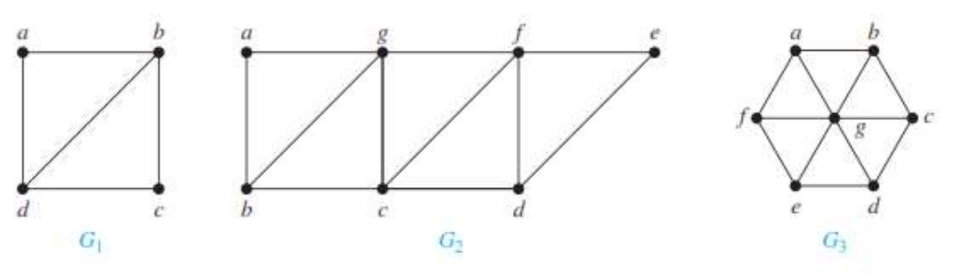
Example: Which of the undirected graphs have an Euler circuit? Of those that do not, which have an Euler path?



Solution: Each vertex of graph G1 has an even degree, so it has an Euler circuit for example, a, e, c, d, e, b, a. Neither of the graphs G2 or G3 has an Euler circuit. However, G3 has an Euler path as it has exactly two vertices (a, b) of odd degree, namely, a, c, d, e, b, d, a, b. G2 does not have an Euler path.

Euler paths & circuits

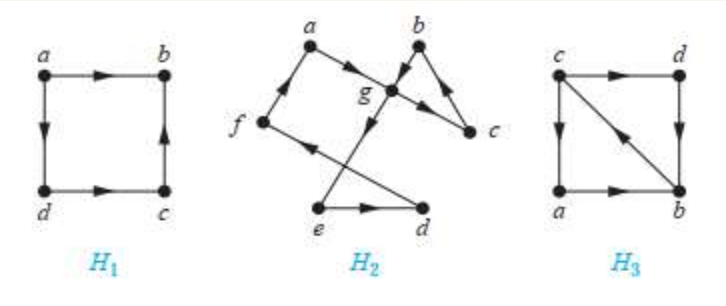
Example: Which of the undirected graphs have an Euler circuit? Of those that do not, which have an Euler path?



Solution: All three graphs have no Euler circuit. G1 contains exactly two vertices of odd degree, namely, b and d. Hence, it has an Euler path that must have b and d as its endpoints. One such Euler path is d, a, b, c, d, b. Similarly, G2 has exactly two vertices of odd degrees, namely, b and d. So it has an Euler path that must have b and d as endpoints. One such Euler path is b, a, g, f, e, d, c, g, b, c, f, d. G3 has no Euler path because it has six vertices of odd degrees.

Euler paths & circuits

Example: Which of the directed graphs have an Euler circuit? Of those that do not, which have an Euler path?



Solution: The graph H2 has an Euler circuit, for example, a, g, c, b, g, e, d, f, a. Neither H1 nor H3 has an Euler circuit. H3 has an Euler path, namely, c, a, b, c, d, b, but H1 does not.

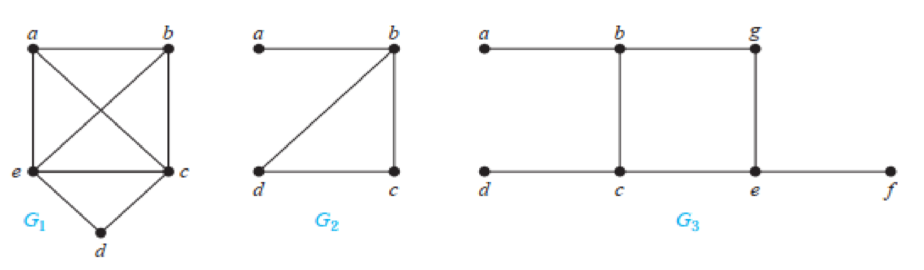
Hamilton paths

A simple path in a graph G that passes through every vertex exactly once is called a *Hamilton path*, and a simple circuit in a graph G that passes through every vertex exactly once is called a *Hamilton circuit*. That is, the simple path $x_0, x_1, \ldots, x_{n-1}, x_n$ in the graph $G = (V, E)$ is a Hamilton path if $V = \{x_0, x_1, \ldots, x_{n-1}, x_n\}$ and $x_i \neq x_j$ for $0 \leq i < j \leq n$, and the simple circuit $x_0, x_1, \ldots, x_{n-1}, x_n, x_0$ (with $n > 0$) is a Hamilton circuit if $x_0, x_1, \ldots, x_{n-1}, x_n$ is a Hamilton path.

[**Spr'16**]

Hamilton paths

Example: Which of the simple graphs have a Hamilton circuit or, if not, a Hamilton path?



Solution: G1 has a Hamilton circuit: a, b, c, d, e, a. There is no Hamilton circuit in G2 (this can be seen by noting that any circuit containing every vertex must contain the edge {a, b} twice), but G2 does have a Hamilton path, namely, a, b, c, d. G3 has neither a Hamilton circuit nor a Hamilton path, because any path containing all vertices must contain one of the edges {a, b}, {e, f }, and {c, d} more than once.